period, preferably on yearly basis. The formula used to calculate this mortality rate for a given period is

$$CDR = \frac{D}{P} \times 1000$$

where

*D*: Number of deaths in the population or community during the given period, and

*P*: Number of persons in the population or community during the given period.

## Example 8.1

There were 15,000 persons living in a village during a period and the number of persons dead during the same period was 98.

Then, the *CDR* of the village can be calculated from

$$CDR = \frac{D}{P} \times 1000$$

as

$$CDR = \frac{98}{15000} \times 1000 = 6.53 \text{ } \textit{per thousand.}$$

In some cases, information about the population may be provided in such a manner that people in the population are grouped according to their age into mutually exclusive and exhaustive age groups. The *CDR* can also be calculated based on such kind of information.

## Example 8.2

People living in a town are grouped according to their age into five groups. The number of persons lived during a calendar year and the number of deaths recorded during the same period are as follows:

| Age Group (in years) | 0-10 | 10-30 | 30-50 | 50-70 | 70 and above |
|---|---|---|---|---|---|
| No. of Persons | 5,000 | 10,000 | 15,000 | 10,000 | 2,000 |
| No. of Deaths | 125 | 30 | 30 | 200 | 1,000 |

Calculate crude death rate of the town.

***Solution:***

The total number of deaths occurred in the town, irrespective of the age, is 1385 and the population size is 42,000. Therefore, the *CDR* of the town can be calculated as

$$CDR = \frac{1385}{42000} \times 1000 = 32.98$$

Thus, the Crude Death Rate of the town is 32.98 *per thousand*.



12th_Statistics_EM_Unit_8.indd 212 1/28/2020 6:38:10 PM